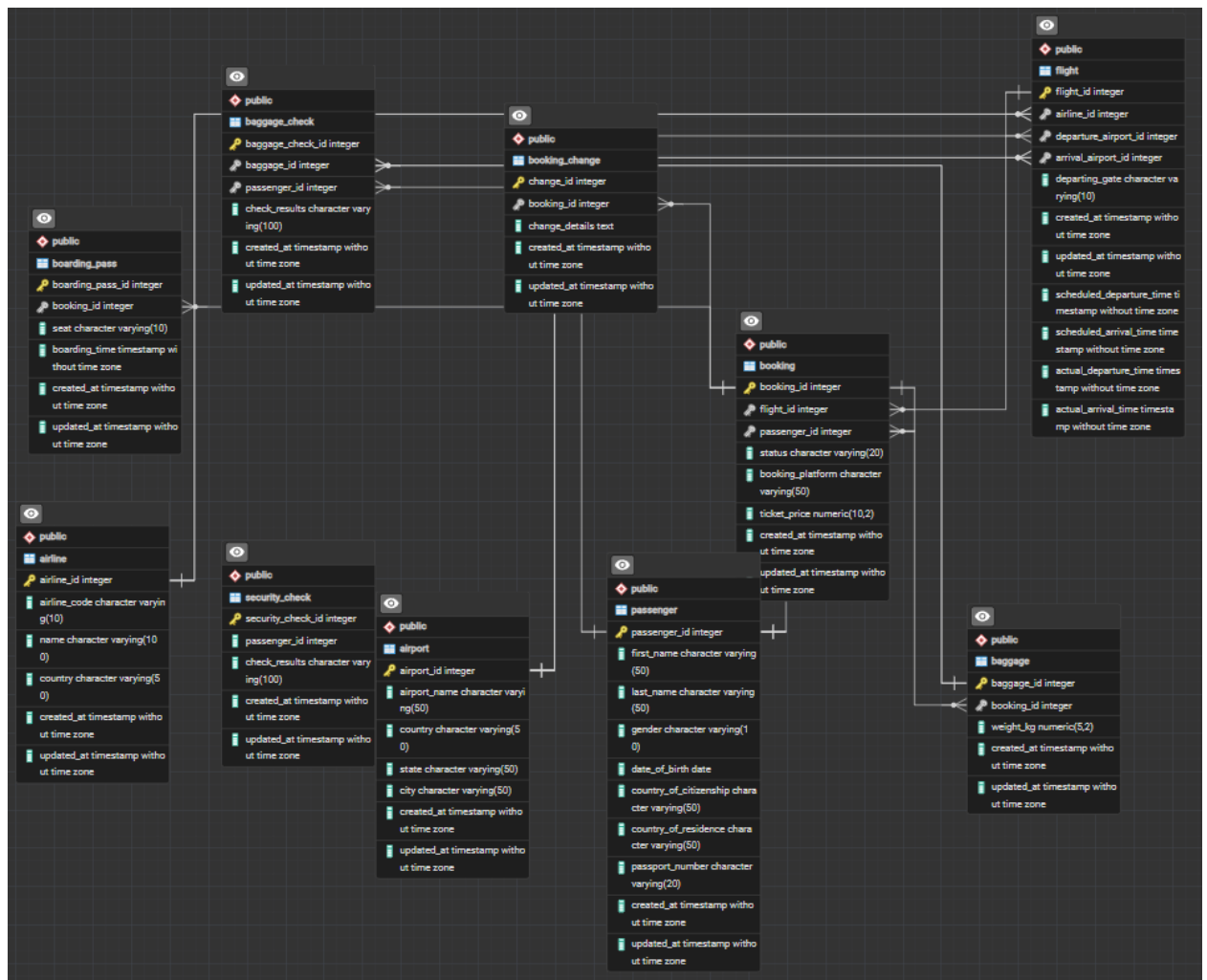




Laboratory Work #1: ERD Diagram for International Airport Database

1. Solution Description

The primary goal of this system is to manage international airport operations, flight scheduling, passenger bookings, boarding processes, baggage processing, and security checks.

2. Legend and Symbols Notation

Entities and Tables are represented as boxes containing attributes and entity names. Primary Key (PK) is a unique identifier for each record in a table. Foreign Key (FK) is an attribute that creates a relationship pointing to a Primary Key in another table. Relationships (1-M) are lines connecting entities showing One-to-Many cardinalities and referential integrity constraints.

3. Entities and Attributes Description

1. airport

Stores geographic and general information about airports. Attributes include airport_id (PK, Integer) as a unique identifier of the airport, airport_name (VarChar) for the full name of the airport, country, state, city (VarChar) for airport location details, and created_at, updated_at (Timestamp) for record metadata.

2. airline

Contains details about commercial air carriers operating at the airport. Attributes include **airline_id** (PK, Integer) as a unique identifier of the airline, **airline_code** (VarChar) for IATA/ICAO short code, **name** (VarChar) for the official name of the airline, **country** (VarChar) for the country of registration, and **created_at**, **updated_at** (Timestamp) for record metadata.

3. flight

Tracks scheduled and actual operational flights. Attributes include **flight_id** (PK, Integer) as a unique flight instance ID, **airline_id** (FK, Integer) for the operating airline, **departure_airport_id** (FK, Integer) for the origin airport, **arrival_airport_id** (FK, Integer) for the destination airport, **departing_gate**, **arriving_gate** (VarChar) for terminal gates, **scheduled_departure_time**, **scheduled_arrival_time** (Timestamp) for scheduled timetable, **actual_departure_time**, **actual_arrival_time** (Timestamp) for actual flight times, and **created_at**, **updated_at** (Timestamp) for record metadata.

4. passenger

Stores personal and identity verification details of passengers. Attributes include **passenger_id** (PK, Integer) as a unique passenger profile ID, **first_name**, **last_name** (VarChar) for the passenger name, **gender** (VarChar) for gender identification, **date_of_birth** (Date) for birth date, **country_of_citizenship**, **country_of_residence** (VarChar) for nationality details, **passport_number** (VarChar) for identification document number, and **created_at**, **updated_at** (Timestamp) for record metadata.

5. booking

Records passenger flight reservations and payment transactions. Attributes include **booking_id** (PK, Integer) as a unique reservation ID, **flight_id** (FK, Integer) for the associated flight, **passenger_id** (FK, Integer) for the associated passenger, **status** (VarChar) for booking status (e.g., Confirmed, Cancelled), **booking_platform** (VarChar) for booking source/website, **ticket_price** (Numeric) for price of the ticket, and **created_at**, **updated_at** (Timestamp) for record metadata.

6. booking_change

Tracks historical changes and modifications made to a booking. Attributes include **change_id** (PK, Integer) as a unique change record ID, **booking_id** (FK, Integer) for target booking reference, **change_details** (Text) for details regarding the modification, and **created_at**, **updated_at** (Timestamp) for record metadata.

7. boarding_pass

Details boarding passes issued to passengers after check-in. Attributes include **boarding_pass_id** (PK, Integer) as a unique boarding pass ID, **booking_id** (FK, Integer) for the associated booking, **seat** (VarChar) for assigned seat number,

boarding_time (Timestamp) for time of boarding, and **created_at, updated_at (Timestamp)** for record metadata.

8. baggage

Registers registered luggage items linked to passenger bookings. Attributes include **baggage_id (PK, Integer)** as a unique luggage tag ID, **booking_id (FK, Integer)** for the associated booking, **weight_kg (Numeric)** for weight in kilograms, and **created_at, updated_at (Timestamp)** for record metadata.

9. baggage_check

Logs baggage screening and inspection outcomes at the airport. Attributes include **baggage_check_id (PK, Integer)** as a unique check ID, **baggage_id (FK, Integer)** for checked luggage, **passenger_id (FK, Integer)** for luggage owner, **check_results (VarChar)** for inspection status (e.g., Passed, Flagged), and **created_at, updated_at (Timestamp)** for record metadata.

10. security_check

Logs individual passenger security control checkpoints. Attributes include **security_check_id (PK, Integer)** as a unique security log ID, **passenger_id (FK, Integer)** for the screened passenger, **check_results (VarChar)** for security check outcome, and **created_at, updated_at (Timestamp)** for record metadata.

4. Relationships and Cardinality Constraints

The relationship from **airline** to **flight** (1:N) represents that one airline operates multiple scheduled flights. The relationship from **airport** to **flight** (1:N) represents that an airport serves as the departure/arrival point for many flights. The relationship from **flight** to **booking** (1:N) represents that one flight can have multiple passenger bookings. The relationship from **passenger** to **booking** (1:N) represents that one passenger can make multiple flight bookings. The relationship from **booking** to **booking_change** (1:N) represents that a single booking can undergo several changes. The relationship from **booking** to **boarding_pass** (1:1 / 1:N) represents that a valid booking leads to issuing a boarding pass. The relationship from **booking** to **baggage** (1:N) represents that one booking can include multiple registered pieces of baggage. The relationship from **baggage** to **baggage_check** (1:N) represents that each baggage item undergoes airport baggage checks. The relationship from **passenger** to **baggage_check** (1:N) links passenger ownership to baggage check logs. The relationship from **passenger** to **security_check** (1:N) represents that a passenger passes through airport security checks.